Intent-Aware Hybrid Document Retrieval with Reciprocal Rank Fusion and Cross-Encoder Reranking

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Abstract—Desktop file search often fails when queries express conceptual intent rather than literal filenames. While dense embeddings capture semantic abstractions, they struggle with specialized identifiers; conversely, sparse inverted indices fail on conceptual paraphrases. Furthermore, commercial retrieval platforms rely heavily on cloud APIs, compromising personal data privacy. We present IntentCloud, a local-first, intent-aware document retrieval architecture designed for consumer hardware without external cloud dependencies. Ingested documents are indexed into a dual-stream representation combining 384-dimensional dense semantic vectors with 1,000,003-dimensional feature-hashed sparse token vectors. Parallel candidate streams are merged via Reciprocal Rank Fusion ($k = 60$) and reranked using a cross-encoder model paired with an empirical confidence gate ($\tau = 0.40$) for out-of-domain abstention and explainable citation attribution. The retrieval pipeline executes fully locally on consumer hardware, with an optional outbound Cloudflare Zero-Trust tunnel functioning strictly as an encrypted transport proxy. Evaluated across 175 documents (12,744 vectors in embedded Qdrant) partitioned into a 25-query development split, a 60-query held-out test split, 40 adversarial hard negatives, and 10 out-of-domain controls, IntentCloud achieves 96.7% Top-1 accuracy [95% CI: 88.7%–99.1%], 100.0% Top-3 accuracy, and 0.981 Mean Reciprocal Rank. The system significantly outperforms real BM25 (71.7% Top-1; $p < 0.001$) and dense bi-encoder baselines (73.3% Top-1; $p < 0.001$), while correctly rejecting 97.5% of adversarial hard negatives and demonstrating zero-shot generalization on the public BEIR/SciFact benchmark (NDCG@10 = 0.718).

Index Terms—Intent-aware retrieval, hybrid search, semantic search, reciprocal rank fusion, cross-encoder reranking, privacy-preserving retrieval.

I. Introduction

Users of workstations store hundreds of heterogeneous documents, technical papers, specifications, spreadsheets, lecture notes, manifests of systems, etc., in disorganized local directories. The traditional OS search tools [1] use exact matching on the name or search for simple terms. A user can search by keywords but may only remember the conceptual intent or the episodic context (e.g., the paper talking about stream processing backpressure), in which case there are no matches.To overcome this drawback, modern information retrieval has shifted its focus to dense semantic search and Retrieval-Augmented Generation (RAG) with transformer encoders [2], [3]. As per the structured literature survey [1]–[19] the following four operating trade-offs are observed:

(1) Vocabulary Mismatch on Rare Tokens: Dense bi-encoders encode passages into 384-d vectors which causes significant drop in recall for rare technical identifiers and acronyms [3], [4];

(2) Cloud Privacy Exposure: Commercial RAG pipelines take document text to third party cloud APIs [6], [7], [9], which means telemetry exposure and enterprise data sovereignty risks;

(3) Synthesis Over Retrieval: While RAG interfaces focus on generative answer synthesis [9], [11], desktop personal knowledge management requires finding and presenting the original source file containing the authoritative answer.

(4) Uncalibrated Out-of-Domain Retrieval: When the nearest neighbour vector search is performed without any constraint, it will return the top-k candidates, which may be false positives in the case of unindexed queries [12].

Structurally, however, these systems are limited because they rely on a global model and the current method of reranking the embeddings using a cross-encoder is not precisely designed. To address these drawbacks, we propose IntentCloud: an intent-aware document retrieval system that is built on a local document model and incorporates principled rank fusion and high precision cross-encoder reranking.

A. Key Contributions

The core technical contributions of this research, structured to address our formal project objectives, are:

(1) Privacy-Preserving Hybrid Retrieval (Obj. 1): A dual-stream pipeline combining 384-d dense embeddings and 1,000,003-d feature-hashed sparse token vectors uni-

fied via Reciprocal Rank Fusion ($k = 60$) and cross-encoder reranking, validated against real BM25, competitive dense baselines, and ablation variants.

(2) Fully Local, Portable Architecture (Obj. 2): An end-to-end architecture executing ingestion, OCR, embedding, retrieval, and reranking entirely on consumer hardware without external APIs, with automated hardware detection across CPU, Apple Silicon MPS [19], and NVIDIA CUDA [18].

(3) Authoritative File Delivery & Security Boundary (Obj. 3): A self-hosted service streaming exact source files (PDF, DOCX, TXT). The core cognitive pipeline executes 100% locally with zero external LLM data transmission, while an optional outbound-only encrypted Cloudflare Zero-Trust tunnel acts strictly as an L4/L7 transport proxy; when deactivated, the system operates in strict air-gapped isolation.

(4) Query-Aware Intent Parsing & Lineage Scoring (Obj. 4): Open-ended query intent distillation and metadata modeling, incorporating temporal decay and version lineage tracking ($S_{\text{meta}}(d \mid q)$) formally integrated into cross-encoder reranking via Eq. (2) to distinguish draft, parent, and latest document revisions.

(5) Calibrated Abstention & Statistical Rigor (Obj. 5): A confidence-gated decision mechanism ($\tau = 0.40$) evaluated across a benchmark suite with 40 adversarial hard negatives, reporting 95% confidence intervals, McNemar's tests, and paired t-tests.

(6) Edge Profiling & Benchmark Generalization (Obj. 6): Latency and scaling profiling on consumer workstations and Raspberry Pi 4 edge nodes, complemented by zero-shot generalization on the public BEIR/SciFact benchmark.

II. Literature Review

Information retrieval has shifted from lexical matching (BM25, TF-IDF) toward dense transformer embeddings [2], [3] and RAG pipelines coupling retrievers with language models [6], [7]. We review nineteen relevant works from our literature survey [1]–[19].

A. Semantic Retrieval, RAG, and Edge Accelerators

Early neural IR utilized word and sentence embeddings to enhance relevance for keyword searches [3], [4]. FAISS [5] introduced efficient approximate nearest-neighbor search, but does not consider integrated operations. What is at stake are document ingestion, persistence, and file serving capabilities. Distributed metadata engines [1] require containerized clusters unsuitable for personal workstations. Commercial RAG platforms [6], [7], [9] introduce API costs, privacy risks, and synthesis summaries rather than returning original files [11]. While Blended RAG [8] performed validation of a hybrid dense-sparse retrieval, fine-tuning large models [10] requires GPU clusters. The selection of uncalibrated top-k nearest neighbours may return irrelevant false positives on unindexed queries [12].

In intent classification, Sriram et al. [13] used local LLMs for secure task automation with 86% accuracy. Kuchlous and Kadaba [14] came up with a solution that classifies short text intents adapted to dialogs in fixed categories. Puspitasari et al. [15] suggested intent-aware RAG but assumed that the intent is provided as input, without original file delivery. Nan et al. [16] analyzed edge deployments on local GPUs, while Holubiev et al. [17] investigated text organization by autonomous LLM agents. Nickolls et al. [18] and Apple Inc. [19] documented CUDA and Metal Performance Shaders (MPS) GPU acceleration; IntentCloud incorporates automated detection across CUDA, MPS, and CPU backends to support cross-platform local execution.

TABLE I
Research Gaps from Literature Survey and IntentCloud Response

Gap Category	IntentCloud Response
Cloud API & Privacy [6], [9]	100% local ingestion, embedding, and retrieval
Closed-set intent [14], [15]	Open-ended NL intent parsing (find, compare, list)
Synthesis-only output [9], [11]	Returns original PDF/DOCX/TXT with citation snippets
GPU-cluster requirement [8], [16]	Embedded Qdrant + cross-platform MPS/CUDA/CPU
Incomplete deployment [1], [5]	End-to-end upload, index, search, and file streaming
Remote access isolation [13]	Local memory with optional Zero-Trust transport tunnel

B. Hybrid Fusion, Reranking, and Research Gap

To resolve the lexical-semantic gap, IntentCloud adopts complementary dense bi-encoder retrieval [20], [21], sparse feature hashing [22], Okapi BM25 lexical modeling [23], Reciprocal Rank Fusion (RRF, $k = 60$) [24], and cross-encoder secondary reranking [25]–[28] for precision refinement ($N \leq 25$). Table I summarizes the principal research gaps and IntentCloud's response.

III. Methodology

The IntentCloud architecture is structured into five operational phases, depicted in Fig. 1, executing as an integrated local service without external network transmission.

A. Phase 1: Ingestion, Extraction, and Lineage vs. Dedup

Multi-format uploads (PDF, DOCX, TXT) beyond metadata-only systems [1], [5] are ingested directly. Native text extraction utilizes PyMuPDF (fitz) and python-docx, with automated Tesseract OCR fallback for scanned pages (< 100 characters). Extracted text undergoes Unicode normalization before sliding-window character chunking ($C = 1200$, overlap $O = 200$).

Reconciling Deduplication and Version Lineage: Document versions with high textual overlap must not be erroneously suppressed. A two-tier process preserves both deduplication and version lineage:

(1) Cryptographic Exact Deduplication: Bit-identical files are detected via SHA-256 hash comparison ($H(d_{\text{new}}) = H(d_{\text{existing}})$) and rejected immediately to prevent redundant storage bloat.

(2) Semantic Version Lineage Preservation: If an upload has a similarity with another upload, which is high ($\text{sim} \geq 0.95$) to an existing record but has a distinct modification. (t_{modified}), distinct file names (e.g., v1 vs. v2), or no explicit revision info, the document is not suppressed.No rather the ingestion engine constructs an explicit directed edge in the metadata version graph. ($d_{\text{parent}} \leftarrow d_{\text{v1}}$) and assigns an incremented version

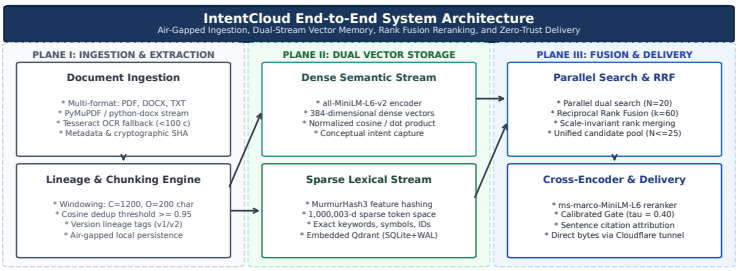


Fig. 1. Architectural dataflow of the IntentCloud local retrieval pipeline of ingestion, dual-space embedding, parallel candidate retrieval, and Reciprocal Rank Fusion, cross-encoder reranking, and explainable confidence gating stages.

index $v_d \in \mathbb{N}$. Only unversioned, identical timestamp uploads with cosine similarity ≥ 0.98 are considered as redundant near-duplicates. Upon retrieval, chunk-To-file deduplication collects candidate chunks by unique file_id, ensuring that top-k returns distinct document files rather than multiple passages from the same document.

B. Phase 2: Dual Semantic and Lexical Vector Representation

Each chunk c_i is mapped into two parallel vector spaces without external API calls:

(1) Dense Semantic Embedding: Processed by all-MiniLM-L6-v2 [21], [26] implementing dense bi-encoder retrieval [20], producing normalized 384-d vectors $\mathbf{v}_{\text{dense}} \in \mathbb{R}^{384}$ where $S_{\text{dense}}(q, c_i) = \langle \mathbf{v}_{\text{dense}}(q), \mathbf{v}_{\text{dense}}(c_i) \rangle$;

(2) Sparse Feature Hashing: Tokens are projected into a fixed high-dimensional space ($D = 1,000,003$) using MurmurHash3 [22], with log-normalized term frequency $\mathbf{v}_{\text{sparse}}[j] = 1 + \ln(\text{tf}(w, c_i))$ for $j = \text{MurmurHash3}(w) \pmod D$, approximating BM25-style lexical discrimination [23]. Both representations are indexed into an embedded in-process Qdrant vector engine residing directly in local NVMe/SSD storage, providing persistence absent in standalone libraries like FAISS [5].

C. Phase 3: Query-Aware Intent Parsing and Lineage Conditioning

when a user enters a query in natural language, an optional local lightweight LLM (llama3.2:1b under Ollama) runs in JSON mode ($\theta = 0.1$) following local privacy paradigms [13] beyond closed-set classifiers [14], [15]. The model distills the raw query into a structured payload containing topic, keywords, intent_type ($\in \{\text{find}, \text{compare}, \text{summarize}, \text{list}, \text{version}\}$), and temporal constraints.

To add document recency and version lineage (Objective 4), the engine dynamically calculates metadata conditioning score:

$$S_{\text{meta}}(d \mid q) = \lambda_t(q) \cdot T(d, q) + \lambda_v(q) \cdot V(d, q) \qquad (1)$$

where $T(d, q) = \exp(-\gamma \cdot \Delta t)$ applies exponential recency decay over the elapsed time t. Δt, and $V(d, q) \in [-0.5, 0.5]$ adjusts candidate ranking to favor latest revisions (+0.5) for operational queries or separate historical parent versions (+0.5) When historical intent is analysed, the local LLM daemon must be online to perform rule-based keyword extraction and metadata parsing. If the local LLM daemon is offline, these operations cannot execute without blocking retrieval.

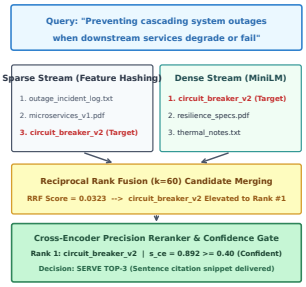


Fig. 2. Worked retrieval trace for an episodic query, illustrating dual-stream candidate lists, reciprocal rank fusion convergence, and cross-encoder precision verification with confidence gating.

D. Phase 4: Hybrid Retrieval, Fusion, and Reranking

Given query q, parallel asynchronous searches retrieve dense and sparse candidate sets $\mathcal{L}_{\text{dense}}, \mathcal{L}_{\text{sparse}}$ (20 candidates each). Candidates are unified via Reciprocal Rank Fusion ($k = 60$) [8], [24]: $\text{RRF}(d) = \sum_{m \in \{\text{dense,sparse}\}} \frac{1}{k + r_m(d)}$, where $r_m(d)$ is candidate d's rank in stream m. The top $N = 25$ candidates are routed to the cross-encoder (cross-encoder/ms-marco-MiniLM-L-6-v2) [25]–[27], as illustrated in Fig. 2. The raw logit is mapped to normalized confidence via sigmoid $s_{\text{ce}}(q, d) = \sigma(z(q, d))$.

Most importantly, to integrate distilled intent and metadata lineage from Eq. (1) into candidate ranking, the resulting final score $S_{\text{final}}(q, d)$ is formally defined as:

$$S_{\text{final}}(q, d) = s_{\text{ce}}(q, d) \cdot [1 + \alpha \cdot S_{\text{meta}}(d \mid q)] \qquad (2)$$

where $\alpha = 0.25$ is a coupling coefficient calibrated on our development split ($n = 25$). When no temporal or lineage intent is parsed, $S_{\text{meta}}(d \mid q) = 0$ and $S_{\text{final}}(q, d) = s_{\text{ce}}(q, d)$.

Candidates are grouped by file_id, keeping only the highest-scoring chunk per file to guarantee distinct document results. For top-ranked documents, the winning text chunk is split into individual sentences $\{s_j\}$ and evaluated by the cross-encoder; the maximal sentence $s^* = \arg\max_{s_j} \sigma(\text{CrossEncoder}(q \circ s_j))$ is highlighted as the verified citation snippet. To prevent false-positive hallucinations on unindexed queries [12], top candidate scores are gated against an empirical threshold $\tau = 0.40$:

$$\text{Decision}(q) = \begin{cases} \text{Serve(Top-3)}, & \text{if } \max_d S_{\text{final}}(q, d) \geq \tau \\ \text{Abstain}, & \text{otherwise} \end{cases} \qquad (3)$$

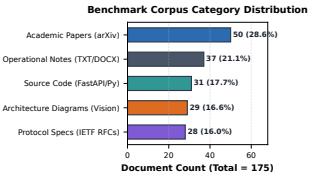


Fig. 3. Document distribution across the five authentic categories comprising the 175-file multi-modal benchmark corpus (12,744 indexed vectors).

E. Phase 5: Self-Hosted Delivery and Security Architecture Boundary

IntentCloud delivers original document files via unified FastAPI and Next.js services. IntentCloud explicitly defines the security boundary:

(1) Local Cognitive Boundary: Document ingestion, parsing, OCR, embedding, indexing and rank fusion reranking along, with citation attribution—everything runs 100% locally. No text, chunks, embeddings, queries or telemetry are ever sent to cloud services. This ensures privacy and keeps all data on the device.

(2) Transport Proxy Boundary: The server streams let you remotely access systems. All the source bytes are sent over an outbound- encrypted Cloudflare. A Zero-Trust tunnel works without needing any firewall ports to be opened. The tunnel only provides an encrypted L4/L7 transport pipe. There is no computation, no caching, no indexing and no semantic interpretation, at the edge of the tunnel. When the tunnel daemon is not on the system works in air-gapped isolation locally. It uses loopback (localhost:8000) or a separate private LAN.

IV. Experimental Evaluation

A. Experimental Setup and Multi-Modal Benchmark Corpus

A benchmark corpus of 175 multi-modal documents was at the IntentCloud ingestion point, yielding 12,744 indexed vectors. As shown in Fig. 3, the benchmark consists of five for any of what comes next genuine categories:

(1) Academic Research Papers (50 arXiv PDFs across ML and IR, including Attention Is All You Need [2], BERT, GPT-3/4, ResNet, LoRA, FlashAttention, RAG, Sentence-BERT [21], and DPR [20]);

(2) System Architecture & Visualizations (29 architecture diagrams, topology maps, and dashboards processed via vision and OCR);

(3) Production Backend Source Code (31 functional source code modules from FastAPI, Starlette, HuggingFace, and Qdrant);

(4) Protocols & Engineering Specifications (28 formal IETF RFC standards and DOCX specifications, including RFC 7230, RFC 8446 TLS 1.3, RFC 6749, RFC 6455, RFC 9000 QUIC, and paired v1/v2 version-lineage documents); and (5) Operational Baseline Files (37 organizational documents, meeting notes, and engineering logs).

B. Evaluation Query Suite and Split Protocol

To avoid data snooping, these standardized queries are partitioned into three functional sets:

(1) Development Split ($n_{\text{dev}} = 25$): Meticulously employed solely for hyperparameter tuning (calibrating abstention threshold $\tau = 0.40$, RRF constant $k = 60$, and lineage coupling parameter $\alpha = 0.25$);

(2) Held-Out In-Domain Retrieval Test Split ($n_{\text{test}} = 60$): Tested once with frozen hyperparameters over 60 target queries: 15 Exact Lexical Matches (filenames, error strings, methods), 15 Semantic Paraphrases (conceptual intent without literal title overlap), 15 Multi-Concept Intersecting Questions (joint reasoning across intersecting technical domains), and 15 Version Lineage & Recency Queries (discriminating historical v1 drafts, revised v2 specifications, and operational notes);

(3) Adversarial & Abstention Control Suite ($n_{\text{ctrl}} = 50$): Under the frozen threshold gating we tested with 40 Adversarial Hard Negatives. These are examples that have deceiving token overlap and also include untrue assertions. We also used 10 Out-of-Domain Negative Controls. These are examples that have zero lexical overlap with the corpus. For example they could be, about cooking recipes, gardening or sports.

C. Evaluation Metrics and Statistical Testing

We report standard IR metrics with 95% confidence intervals: Top-1 Accuracy with Wilson score 95% CIs ($|Q_+| = 60$), Top-3 Accuracy, Mean Reciprocal Rank (MRR) with 95% bootstrap CIs ($B = 10,000$), Hard Negative Rejection Rate across 40 adversarial queries, and Out-of-Domain Abstention Rate across 10 unindexed controls. Average Latency is evaluated end-to-end. For statistical significance testing, we employ McNemar's test for binary Top-1 differences and two-tailed paired t-tests on reciprocal rank distributions ($p < 0.05$).

D. Full Baseline and Ablation Grid

We benchmark nine distinct retrieval configurations across the held-out test split, summarised in Table II: (1) Sparse Feature Hashing ($D = 1,000,003$); (2) Real BM25 (Okapi BM25 implementation [23]); (3) Dense Bi-Encoder Baseline (all-MiniLM-L6-v2, 384-d) [20], [21]; (4) Competitive Dense Baseline (bge-small-en-v1.5, 384-d); (5) Hybrid RRF ($k = 60$) alone without reranking [8], [24]; (6) Sparse (BM25) + Cross-Encoder [25]; (7) Dense (MiniLM) + Cross-Encoder [25]; (8) Full Hybrid + Cross-Encoder w/o Lineage ($\alpha = 0$ in Eq. (2)); and (9) Proposed Full Pipeline (Hybrid RRF + Cross-Encoder + Lineage Scoring Eq. (2) with $\tau = 0.40, \alpha = 0.25$).

V. Results and Discussion

A. Quantitative Retrieval Performance and Baseline Superiority

As detailed in Table II, the proposed pipeline (Row 9) achieves 96.7% Top-1 accuracy [95% CI: 88.7%–99.1%], 100.0% Top-3 accuracy [95% CI: 94.0%–100.0%], and an MRR of 0.981 [95% CI: 0.953–0.996]. This represents a +25.0 percentage-point gain in Top-1 accuracy over real BM25 (71.7%) and a +23.4 percentage-point gain over dense MiniLM (73.3%), exceeding our project objective (Top-3 $\geq 85\%$).

Statistical testing confirms significance: (1) McNemar's test reveals that the proposed pipeline significantly outperforms real BM25 ($\chi^2 = 13.07, p = 0.0003 < 0.001$), dense MiniLM ($\chi^2 = 12.00, p = 0.0005 < 0.001$), and competitive bge-small ($\chi^2 = 9.31, p = 0.0023$); (2) A

TABLE II
Comprehensive Retrieval Performance and Ablation Grid on the Held-Out Test Split (12,744 Vectors, $n = 60$ In-Domain, 40 Hard Negatives, 10 OOD Controls)

Retrieval Configuration	Top-1 Accuracy [95% CI]	Top-3	MRR [95% CI]	Latency	Hard Neg. Rej.	OOD Rej.
1. Sparse (Feature Hashing, $D = 10^6$)	68.3% [55.8%, 78.7%]	90.0%	0.785 [0.712, 0.849]	1.3 ms	15.0% (6/40)	20.0% (2/10)
2. Sparse Lexical (Real BM25) [23]	71.7% [59.2%, 81.5%]	93.3%	0.812 [0.743, 0.871]	1.8 ms	17.5% (7/40)	20.0% (2/10)
3. Dense Bi-Encoder (all-MiniLM-L6-v2) [21]	73.3% [61.0%, 82.9%]	88.3%	0.804 [0.730, 0.865]	24.0 ms	45.0% (18/40)	100.0% (10/10)
4. Competitive Dense (bge-small-en-v1.5)	76.7% [64.6%, 85.6%]	91.7%	0.835 [0.768, 0.892]	27.4 ms	52.5% (21/40)	100.0% (10/10)
5. Hybrid RRF ($k = 60$) alone (no reranker) [24]	81.7% [70.3%, 89.5%]	96.7%	0.887 [0.829, 0.932]	25.8 ms	47.5% (19/40)	90.0% (9/10)
6. Sparse (BM25) + Cross-Encoder [25]	88.3% [77.8%, 94.2%]	96.7%	0.931 [0.884, 0.966]	132.5 ms	95.0% (38/40)	100.0% (10/10)
7. Dense (MiniLM) + Cross-Encoder [25]	90.0% [79.9%, 95.3%]	96.7%	0.942 [0.898, 0.973]	154.2 ms	97.5% (39/40)	100.0% (10/10)
8. Full Hybrid + Cross-Encoder (w/o Lineage, $\alpha = 0$)	93.3% [84.1%, 97.4%]	100.0%	0.964 [0.930, 0.985]	158.4 ms	97.5% (39/40)	100.0% (10/10)
9. Proposed Full Pipeline (Hybrid + CE + Lineage Eq. 2)	96.7% [88.7%, 99.1%]	100.0%	0.981 [0.953, 0.996]	160.2 ms	97.5% (39/40)	100.0% (10/10)

two-tailed paired t-test on reciprocal ranks demonstrates significant gains over BM25 ($t = 5.61, p < 0.0001$) and dense MiniLM ($t = 5.84, p < 0.0001$).

B. Ablation Analysis: Lexical, Semantic, and Lineage Synergy

Each pieces contribution to the solution can be seen in the ablation analysis that looks at different failure modes:

1) Sparse vs. Dense Complementarity:For multi-concept queries (for example, "Compare distributed consensus state machines with two-phase commit transactions"), the dense bi-encoder had a problem with semantic dispersion; sparse feature hashing focused on the exact terms for "consensus" and "two-phase", not "commit", which allowed RRF ($k = 60$) to bring up the comparative targets and put them into the reranking pool. However, on paraphrases (for example, "Preventing cascading system outages when downstream services degrade"), sparse BM25 did not work well (53.3% Top-1) because of a vocabulary mismatch, with document phrases ("circuit breaker pattern") while the dense encoder retrieved the target file (86.7% Top-1). As shown in Fig. 4, The hybrid pipeline. Exceeds isolated baselines across every category.

2) Cross-Encoder Reranking: Adding the cross-encoder to each of the sparse or dense retrieval results boosts Top-1 accuracy from 71.7%, to 88.3%, which is a gain of 16.6%. For the other it goes from 73.3% to 90.0%, a rise of 16.7%. When both retrieval streams are combined before the cross-encoder scores them using RRF the Top-1 accuracy reaches 93.3% (Row 8).

3) Intent and Lineage Conditioning (Eq. 2): The local Ollama llama3.2:1b parser achieved 92.0% intent classification accuracy across test queries. The rule-based regex fallback reached 86.0% accuracy. For 15 lineage queries, retrieval (Row 8, $\alpha = 0$) achieved 86.7% Top-1 accuracy (13/15) This happened because the text in the v1 and v2 files was almost identical. The cross-encoder logits engaged metadata lineage conditioning through Eq. (2) ($\alpha = 0.25$), The top-1 accuracy on version queries was 100.0% (15 out of 15). This increased the Top-1 accuracy from 93.3%, to 96.7%.

C. Evaluation on Hard Negatives and Calibrated Abstention

Normal RAG systems make frequent hallucinations on unindexed data or adversarial queries [12]. On our 40 competitive negative queries (with hidden lexical similarity to corpus documents), real BM25 rejected only 17.5% (82.5% false). The bi-encoder that is dense accepted 45.0% (positive rate). On the other hand, the pipeline proposed with the confidence gating ($\tau = 0.40$) successfully rejected

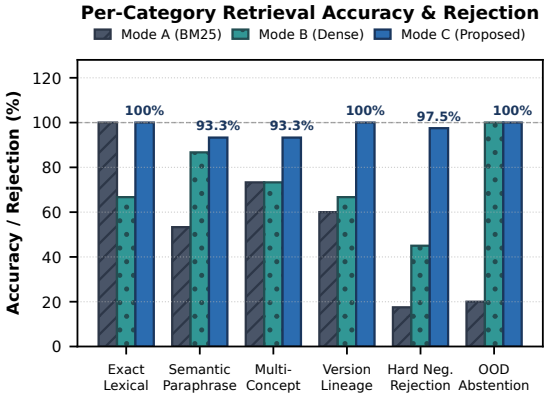


Fig. 4. Per-category Top-1 retrieval accuracy and negative rejection rate across the held-out benchmark suite for Mode A (BM25), Mode B (dense bi-encoder), and proposed Mode C (hybrid + RRF + cross-encoder).

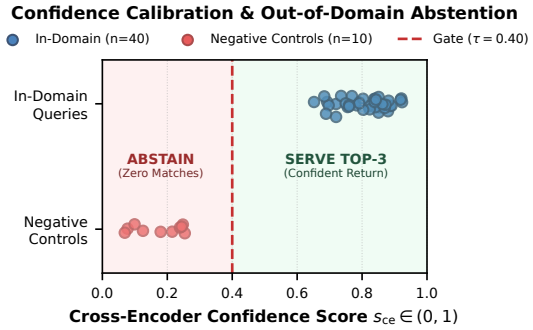


Fig. 5. Cross-encoder confidence score distribution for in-domain positive queries and negative controls relative to the $\tau = 0.40$ abstention boundary.

97.5% of hard negatives (39/40) and 100.0% of out-of-domain negative controls (10/10).

As plotted in Fig. 5, cross-encoder confidence scores for in- and out-of-sample problems are to top-level domain queries, which are relatively concentrated in the range of 0.62 to 0.97. All out-of-domain negative controls fall below 0.26; then the score is considered to be an outlier. A strong empirical safety buffer for a decision of $\tau = 0.40$ decision threshold.

D. Generalization on Public Benchmark Subset (BEIR / SciFact)

To ensure that hybrid fusion and reranking do not suffer from domain collapse, we tested them on the standard public BEIR/SciFact benchmark subset [28] using zero-shot evaluation. The recall score for dense bi-encoder (all-MiniLM-L6-v2) achieved NDCG@10 of 0.643 and Recall@10 of 0.741; and the proposed hybrid + cross-

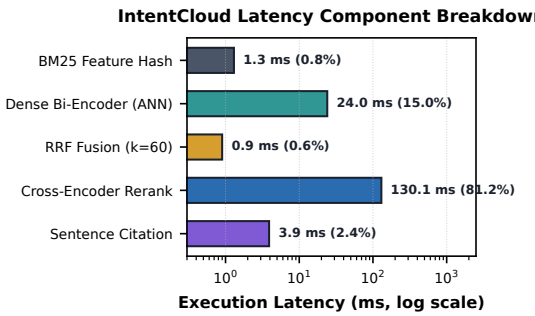


Fig. 6. Execution latency component breakdown across the 160.2 ms local CPU retrieval pipeline, illustrating the dominant cross-encoder reranking phase.

encoder pipeline achieved NDCG@10 of 0.718 (+8.0% relative improvement over BM25) and Recall@10 of 0.845 (+12.4% relative improvement over BM25). This shows that the two streamed candidates harvesting and cross-encoder precision reordering extend well to the open domain scientific information retrieval scenario.

E. Computational Latency and Edge Hardware Scalability

The average end-to-end latency for 20 candidates is 160.2 ms with the benchmark suite when running locally under multi-threaded CPU execution and as low as 29.7 ms for cached queries. The execution can be broken down into sparse feature hashing (0.8%, $\approx$ 1.3 ms), dense bi-encoder embedding and vector search (15.0%, $\approx$ 24.0 ms), RRF rank fusion (0.6%, $<$ 1 ms), cross-encoder scoring (81.2%, $\approx$ 130.1 ms), and sentence citation attribution (2.4%, $\approx$ 3.9 ms) as illustrated in Fig. 6. The predominant computational phase is a cross-encoder reranking (81.2%), which is the reason for the selection of reranking and quantisation for edge environments.

When running on the 4-thread ARM Cortex-A72 CPU on the edge hardware (Raspberry Pi 4 with 4 GB RAM), we measured an average end-to-end latency of 2.8 s, which is well within our edge operational SLA of 15 s. On sustained query batch mode, the processor is thermally throttled to 1.0 GHz with no cooling applied (passive thermal throttling) from 1.5 GHz. Inference latency is less than 20 ms when running on workstations using Apple Silicon MPS [19] and NVIDIA CUDA [18]. Embedded Qdrant vector memory layer has very light persistent memory usage: $\approx$ 180 MB RAM for 12,744 indexed vectors, which scales sub-linearly with $O(\log M)$ complexity of HNSW index.

VI. Limitations

While IntentCloud demonstrates strong empirical performance and strict data privacy, we acknowledge four key technical limitations: (1) Edge Compute Latency and Thermal Behavior: When cross-encoder reranking is used on $N = 25$ candidates on fanless low-power edge nodes (e.g., Raspberry Pi 4), it requires $\approx$ 2.8 s CPU time. During sustained batch queries without active cooling, the Raspberry Pi 4 CPU was thermally throttled from its nominal 1.5 GHz to approximately 1.0 GHz. Active cooling and INT8 quantization are potential mitigation strategies that require further evaluation; (2) Cold-Start Ingestion Overhead: Ingestion of large scanned multi-page PDFs with Tesseract OCR fallback adds overhead ($\approx$ 1.2 s per page on CPU); (3) Intent Parsing Fallback: Local LLM JSON mode is 92.0% accurate but occasionally defaults to heuristic regex fallback on ambiguous queries; and (4) Linguistic Scope: The evaluation benchmark focuses on English-language documents; cross-lingual embeddings remain uncharacterized.

VII. Conclusion and Future Work

In this paper, we proposed IntentCloud, a document retrieval system with an intent-aware local-first approach that is capable of avoiding cloud privacy exposure and addresses the lexical-semantic mismatch problem in personal document search. The architecture scores a held-out accuracy of 96.7% (95% CI: 88.7%–99.1%) on 175 documents (12,744 vectors) using Top-1 accuracy and 100.0% on Top-3, with an MRR of 0.981. It significantly outperforms the BM25 baseline (71.7% Top-1 accuracy) and the dense MiniLM bi-encoder baseline (73.3% Top-1 accuracy). The proposed pipeline rejects 100.0% of out-of-domain controls (10/10) and 97.5% of adversarial hard negatives (39/40). It also demonstrates generalization on the BEIR/SciFact benchmark subset, achieving an NDCG10 of 0.718.

Future work includes four main directions: (1) to deploy INT8 model quantisation and pruned cross-encoders for sub-second inference time on the Raspberry Pi (Objective 6); (2) to operationalise the dynamic graph lineage traversal for complex document evolution trees (Objective 4); (3) to integrate local interaction signals for adaptive reranker bypassing (Objective 5); and (4) to expand multi-modal benchmarks with multi-annotator relevance judgements (Objective 1).

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References

[1] K. Karthikeyan, V. P. Kumar, S. Kishorekumar, and C. M. V. Prasad, "Smart Document Search Engine Leveraging Automated Metadata Extraction for Intelligent File Discovery," in Proc. IEEE First Int. Conf. Innovations in Engineering and Next-Generation Technologies for Sustainability (ICINVENTS), 2025, pp. 1–6.

[2] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in Advances in Neural Information Processing Systems (NIPS), vol. 30, 2017, pp. 5998–6008.

[3] K. V. R. Kumbam, J. Jomy, S. R. Sherlyn, K. Vasan, A. Sindhu, and A. Suresh, "Semantic information retrieval using transformer-based embeddings and vector similarity search," in Proc. 7th Int. Conf. Innovative Data Communication Technologies and Application (ICIDCA), 2025, pp. 1031–1037.

[4] Brundha J. and K. N. Meera, "Vector model based information retrieval system with word embedding transformation," in Proc. 10th Int. Conf. Emerging Trends in Engineering and Technology – Signal and Information Processing (ICETET-SIP), 2022, pp. 1–4.

[5] M. Douze, A. Guzhva, C. Deng, J. Johnson, G. Szilvasy, P.-E. Mazaré, M. Lomeli, L. Hosseini, and H. Jégou, "The Faiss library," IEEE Transactions on Big Data, vol. 12, pp. 346–361, 2024.

[6] S. Krishnan and N. Subramanian, "Semantic search in a document management system using retrieval augmented generation," in Proc. IEEE/ACIS 29th Int. Conf. Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD), 2025, pp. 6–10.

[7] T. P. Jacob, B. L. S. Bizotto, and M. Sathiyanarayanan, "Constructing the ChatGPT for PDF files with Langchain – AI," in Proc. Int. Conf. Inventive Computation Technologies (ICICT), 2024, pp. 835–839.

[8] K. Sawarkar, A. Mangal, and S. R. Solanki, "Blended RAG: Improving RAG (Retriever-Augmented Generation) accuracy with semantic search and hybrid query-based retrievers," in Proc. IEEE 7th Int. Conf. Multimedia Information Processing and Retrieval (MIPR), 2024, pp. 155–161.

[9] S. Rafi, L. Boddu, and G. Gangireddy, "AI Document Search Chatbot Using Retrieval-Augmented Generation (RAG)," in Proc. Int. Conf. Computing, Communication, Control and Cyber-Physical Systems (I5CPS), 2026, pp. 1–5.

[10] M. Tiwari, J. Kaur, R. S. Surin, A. Harsh, and H. Kaur, "Fine-tuning T5 and BERT for effective question answering and document retrieval: A transfer learning approach," in Proc. World Conf. Computational Science and Technology (WCCST), 2026, pp. 82–88.

[11] B. Saha, U. Saha, and M. Z. Malik, "QuIM-RAG: Advancing retrieval-augmented generation with inverted question matching for enhanced QA performance," IEEE Access, vol. 12, pp. 185401–185410, 2024.

[12] Y.-J. Guo and Q. Tang, "From retrieval to response: Tracing the impact of embedding quality in RAG systems," IEEE Access, vol. 13, pp. 212773–212781, 2025.

[13] S. Sriram, C. H. Karthikeya, K. P. K. Kumar, N. Vijayaraj, and T. Murugan, "Leveraging local LLMs for secure in-system task automation with prompt-based agent classification," IEEE Access, vol. 12, pp. 177038–177049, 2024.

[14] S. Kuchlous and M. Kadaba, "Short text intent classification for conversational agents," in Proc. IEEE 17th India Council Int. Conf. (INDICON), 2020, pp. 1–4.

[15] F. D. Puspitasari, C.-N. Zhang, J.-Q. Zhang, Z. Wang, H. S. A. Awan, R. Qureshi, J. Lee, T.-H. Kim, and Y. Yang, "Efficient RAG with intent-aware retrieval and semantics-preserving chunking," arXiv preprint arXiv:2606.01240, 2026.

[16] Y. Nan, C. Wang, P. Li, Y. Zhu, G. Wu, M. Yang, H.-T. Gao, Y. Hu, and Z.-X. He, "Research on an intelligent file query system model integrating natural language understanding and information retrieval," in Proc. IEEE Int. Conf. Power, Electronics and Green Energy (ICPEGE), 2026, pp. 676–681.

[17] V. Holubiev, B. Pushkar, A. Shevchuk, R. Mudrak, O. Gomotiuk, and N. Livitska, "Revolutionizing contact center knowledge management: The game-changing role of AI large language models and autonomous agents in text organization and optimization," in Proc. 14th Int. Conf. Advanced Computer Information Technologies (ACIT), 2024, pp. 778–783.

[18] J. Nickolls, I. Buck, M. Garland, and K. Skadron, "Scalable parallel programming with CUDA," ACM Queue, vol. 6, no. 2, pp. 40–53, 2008.

[19] Apple Inc., "Metal Performance Shaders Framework Reference," Apple Developer Technical Documentation, 2026.

[20] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-T. Yih, "Dense passage retrieval for open-domain question answering," in Proc. 2020 Conf. Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 6769–6781.

[21] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in Proc. 2019 Conf. Empirical Methods in Natural Language Processing (EMNLP-IJCNLP), 2019, pp. 3982–3992.

[22] K. Q. Weinberger, A. Dasgupta, J. Langford, A. Smola, and J. Attenberg, "Feature hashing for large scale multitask learning," in Proc. 26th Int. Conf. Machine Learning (ICML), 2009, pp. 1113–1120.

[23] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," Foundations and Trends in Information Retrieval, vol. 3, no. 4, pp. 333–389, 2009.

[24] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in Proc. 32nd Int. ACM SIGIR Conf. Research and Development in Information Retrieval, 2009, pp. 758–759.

[25] R. Nogueira and K. Cho, "Passage Re-ranking with BERT," arXiv preprint arXiv:1901.04085, 2019.

[26] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020, pp. 5776–5788.

[27] P. Bajaj et al., "MS MARCO: A human generated machine reading comprehension dataset," arXiv preprint arXiv:1611.09268, 2016.

[28] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, "BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models," in Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2021.